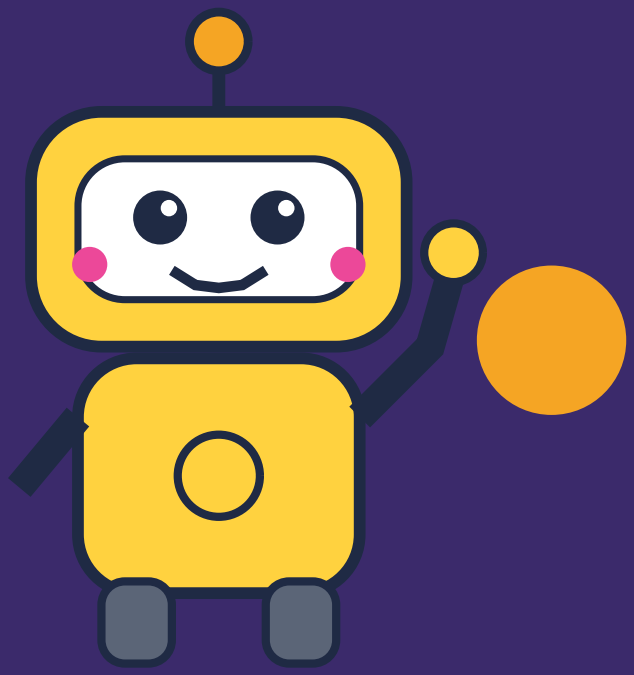
ASSESSMENT BOOK

Think Like a Coder

Chapter assessments for Class 7

12 one-page assessments, one for every chapter
10 marks each: choose, apply and think like a coder
Answers, marking guides and a class record sheet



For CBSE's Computational Thinking and AI curriculum, 2026-27

Goes with Think Like a Coder, Class 7 · Photocopiable for classroom use

SAMPLE EDITION · DRAFT

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Using these assessments

This book has one assessment for every chapter of *Think Like a Coder*, Class 7. Each assessment fits on one page, takes about 30 minutes and is marked out of 10. Use it after the chapter's last period, or as a revision check later in the term. All questions are new: they test the same skills as the workbook, not the same questions.

Section	What it checks	Questions	Marks
A	Knowing and understanding the chapter's ideas	3 multiple choice	3 x 1
B	Applying the idea to a short problem, with working	2 short answers	2 x 2
C	Thinking like a coder: write, trace, fix or explain a program or plan	1 task	1 x 3

Turning a score into a journal level

Score	Journal level	What it usually means
9-10	Extending	Solves new problems and explains the reasoning
7-8	Secure	Applies the idea correctly on their own
4-6	Developing	Understands with prompts; makes some slips
0-3	Beginning	Needs more practice with the chapter's idea

✓ Good practice

- ✓ Read the questions aloud for children who find reading hard: we are testing thinking, not reading.
- ✓ Give full marks for any correct method, even if it differs from the answer key; the marking guides list what to look for.
- ✓ Give part marks in Sections B and C for correct working, even when the final answer has a slip.
- ✓ Record the score on the class record sheet, and use it with your observations in the journal, not instead of them.
- ✓ Go over the test with the class: the most common mistakes are a lesson plan for the next period.

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Assessment 1 Large Numbers Around Us

Coder skill: Compare, round and estimate big numbers step by step · Level C4 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 1 crore = ... lakh [1 mark]

a) 10 **b)** 100 **c)** 1,000

2 1 million = ... lakh [1 mark]

a) 1 **b)** 10 **c)** 100

3 Which number is bigger? [1 mark]

a) 9,99,999 **b)** 10,00,001 **c)** they are equal

Section B Answer in a few words or steps — 4 marks

4 Write 3,05,40,000 in words (Indian system). Then write it with international commas. [2 marks]

5 Round 47,862 and 31,240 to the nearest thousand, and use them to estimate 47,862 + 31,240. [2 marks]

Section C Think like a coder — 3 marks

6 Write the steps a computer uses to compare two big numbers. Use them to compare 7,04,56,312 and 70,456,231. [3 marks]

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Assessment 2 Arithmetic Expressions

Coder skill: Order of operations and expression trees · Level C4 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 10 - 2 x 3 = ... [1 mark]

a) 24 **b)** 4 **c)** 18

2 (10 - 2) x 3 = ... [1 mark]

a) 24 **b)** 4 **c)** 18

3 In an expression tree, the operation at the top is done ... [1 mark]

a) first **b)** last **c)** never

Section B Answer in a few words or steps — 4 marks

4 Work out 20 ÷ 4 + 3 x (6 - 2). Show the order. [2 marks]

5 Put brackets in 2 + 3 x 4 so that the answer is 20. [2 marks]

Section C Think like a coder — 3 marks

6 Draw the expression tree for (7 + 5) x (9 - 4). Then work it out from the bottom up. [3 marks]

How did I do? Colour a face.

I can finish this test.	I can show my thinking.	I can check my answers.
		

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Assessment 3 A Peek Beyond the Point

Coder skill: Decimals, binary numbers and punched-card sorting · Level C4 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 In 3.47, the 7 means 7 ... [1 mark]

a) tenths **b)** hundredths **c)** ones

2 The binary number 101 is ... in decimal. [1 mark]

a) 5 **b)** 3 **c)** 101

3 How many different patterns can 5 bits make? [1 mark]

a) 10 **b)** 25 **c)** 32

Section B Answer in a few words or steps — 4 marks

4 Write 19 in binary. [2 marks]

5 Change the binary number 11010 into decimal. [2 marks]

Section C Think like a coder — 3 marks

6 A school has 50 clubs and wants a binary code for each. How many bits are needed? Explain. Write the code for club 37. [3 marks]

How did I do? Colour a face.

I can finish this test.	I can show my thinking.	I can check my answers.
		

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Assessment 4 Expressions using Letter-Numbers

Coder skill: Variables and formulas for growing patterns · Level C4 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 The perimeter of a square with side s is ... [1 mark]

a) s + 4 **b)** 4s **c)** s x s

2 If n = 5, then 3n + 1 = ... [1 mark]

a) 16 **b)** 36 **c)** 9

3 4x + 3x = ... [1 mark]

a) 7x **b)** 12x **c)** 7

Section B Answer in a few words or steps — 4 marks

4 A pattern has 6, 11, 16, 21, ... in steps 1, 2, 3, 4. Write a formula for step n, and find step 20. [2 marks]

5 Simplify 5a + 2b + 3a - b. [2 marks]

Section C Think like a coder — 3 marks

6 A program says: set tiles to 2 x n + 4. Find tiles for n = 1, 2 and 3. Which n gives 50 tiles? [3 marks]

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Assessment 5 Parallel and Intersecting Lines

Coder skill: Angle rules and tracing drawing programs · Level C5 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 Two lines cross and one angle is 40°. The vertically opposite angle is ... [1 mark]

a) 40° **b)** 140° **c)** 50°

2 Angles on a straight line add up to ... [1 mark]

a) 90° **b)** 180° **c)** 360°

3 A transversal crosses two parallel lines. Its 8 angles come in how many sizes? [1 mark]

a) one **b)** two **c)** eight

Section B Answer in a few words or steps — 4 marks

4 Two lines cross and one angle is 72°. Find the other three angles. [2 marks]

5 A turtle faces 0°. Program: forward 50, turn right 60°, forward 30, turn left 60°, forward 50. Are the first and last lines parallel? Why? [2 marks]

Section C Think like a coder — 3 marks

6 A transversal crosses two parallel lines and one angle is 115°. How many of the 8 angles are 115°, and how many are 65°? Explain. [3 marks]

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Assessment 6 Number Play

Coder skill: True or false: list every case and filter with clues · Level C5 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 With 3 lights, each ON or OFF, how many different cards are there? [1 mark]

a) 6 **b)** 8 **c)** 9

2 With 4 lights, how many cards are there? [1 mark]

a) 8 **b)** 12 **c)** 16

3 Filtering cases means ... [1 mark]

a) removing the cases that break a clue **b)** adding more cards

c) guessing

Section B Answer in a few words or steps — 4 marks

4 Lights R, G and B. List every case where exactly two lights are ON. [2 marks]

5 Work out 83 - 38 and 61 - 16. What do you notice? [2 marks]

Section C Think like a coder — 3 marks

6 Explain why a 2-digit number minus its reverse is always a multiple of 9. (Hint: write the number as 10a + b.) [3 marks]

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Assessment 7 A Tale of Three Intersecting Lines

Coder skill: Triangle rules as checks; debugging a function · Level C5 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 The angles of a triangle add up to ... [1 mark]

a) 90° **b)** 180° **c)** 360°

2 Which lengths can make a triangle? [1 mark]

a) 2, 3, 6 **b)** 4, 5, 8 **c)** 1, 1, 3

3 A triangle with exactly 2 equal sides is ... [1 mark]

a) equilateral **b)** isosceles **c)** scalene

Section B Answer in a few words or steps — 4 marks

4 Two angles of a triangle are 48° and 67°. Find the third angle. Is the triangle acute, right or obtuse? [2 marks]

5 Can 5 cm, 5 cm and 10 cm make a triangle? Why? [2 marks]

Section C Think like a coder — 3 marks

6 A function says: isTriangle(a, b, c): if a + b > c then say 'yes' else say 'no'. Give a test that shows the bug. Then fix the function. [3 marks]

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Assessment 8 Working with Fractions

Coder skill: Fraction operations and working backwards · Level C4 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 3/4 of 48 = ... [1 mark]

a) 12 **b)** 36 **c)** 64

2 2/3 x 3/5 = ... [1 mark]

a) 2/5 **b)** 5/8 **c)** 1

3 The reciprocal of 3/4 is ... [1 mark]

a) 4/3 **b)** 3/4 **c)** 1/4

Section B Answer in a few words or steps — 4 marks

4 How many 2/3-litre bottles can you fill from 6 litres? Show the division. [2 marks]

5 I think of a number, multiply it by 4, subtract 5 and get 23. Work backwards to find it. [2 marks]

Section C Think like a coder — 3 marks

6 Ravi spent 1/2 of his money on books, then 1/3 of what was left on lunch. He had ₹40 left. How much did he start with? Work backwards. [3 marks]

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Assessment AI1 AI Domains and Applications

Coder skill: Classify, predict a number, or find groups · Level C5 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 Recognising faces in photos belongs to ... [1 mark]

a) Data Science **b)** Computer Vision

c) Natural Language Processing

2 A voice assistant understanding speech belongs to ... [1 mark]

a) Computer Vision **b)** Natural Language Processing

c) Data Science

3 Predicting tomorrow's temperature is ... [1 mark]

a) classification **b)** regression **c)** clustering

Section B Answer in a few words or steps — 4 marks

4 Classification, regression or clustering? (a) Sorting emails into spam and not spam. (b) Grouping songs that sound alike, with no labels. [2 marks]

5 Practice hours and marks: 1 hour 40, 2 hours 45, 3 hours 50. Predict the marks for 5 hours. Which kind of task is this? [2 marks]

Section C Think like a coder — 3 marks

6 A dataset has 1,000 labelled photos. Explain how you would split it into training, validation and test parts, and why the test part must be unseen. [3 marks]

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Assessment AI2 AI in Industries

Coder skill: AI in health, education, transport and communication · Level C5 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 AI that reads X-rays to spot illness is used in ... [1 mark]

a) healthcare **b)** transport **c)** communication

2 A route app chooses the route with the shortest ... [1 mark]

a) distance **b)** travel time **c)** name

3 Who is responsible for the final diagnosis? [1 mark]

a) the AI **b)** the doctor

c) the hospital computer

Section B Answer in a few words or steps — 4 marks

4 Give one use of AI in education and one in communication. [2 marks]

5 Route A is 12 km and should take 30 minutes. Route B is 15 km and should take 22 minutes. Which does the app choose? Why? [2 marks]

Section C Think like a coder — 3 marks

6 An AI system suggests which hospital patients need urgent care. Give one benefit, one risk, and one way to keep humans responsible. [3 marks]

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Assessment AI3 Data Visualisation and Analysis

Coder skill: Choose bar, line or pie charts and read them honestly · Level C5 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 Change over time is best shown with a ... [1 mark]

a) bar chart **b)** line graph **c)** pie chart

2 In a pie chart, the whole circle is ... [1 mark]

a) 100° **b)** 180° **c)** 360°

3 A photo is ... data. [1 mark]

a) structured **b)** unstructured

Section B Answer in a few words or steps — 4 marks

4 60 students: 30 play cricket, 15 football and 15 hockey. Find the angle of each slice in a pie chart. [2 marks]

5 A bar chart shows marks of 92 and 96, but its axis starts at 90, so 96 looks three times as big. Why is this misleading? How do you fix it? [2 marks]

Section C Think like a coder — 3 marks

6 Temperatures Monday to Friday: 28°C, 30°C, 31°C, 29°C, 33°C. Which chart would you use? Describe the trend. Give one rule for an honest chart. [3 marks]

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Assessment AI4 Ethics and AI Bias Awareness

Coder skill: Spot bias in data and keep humans in charge · Level C5 · 30 minutes · 10 marks

Name _____ **Section** _____ **Date** _____ **Score** ___ / 10

Section A Choose the right answer — 3 marks

1 Bias in AI usually comes from ... [1 mark]

a) the data **b)** the electricity **c)** the screen

2 Training data that leaves out some groups shows ... [1 mark]

a) data bias **b)** historical bias

c) measurement bias

3 For important decisions there must be a ... [1 mark]

a) human in the loop **b)** faster computer

c) bigger screen

Section B Answer in a few words or steps — 4 marks

4 A classifier gets 45 of 50 fruits right and 3 of 10 vegetables right. What is its accuracy for each group? [2 marks]

5 A model learns from old hiring records in which almost only men were chosen. Which kind of bias is this? Why? [2 marks]

Section C Think like a coder — 3 marks

6 Face unlock works well for adults but often fails for children. Suggest a likely cause, how to check it, and how to fix it. [3 marks]

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Answers and marking guide

Accept any correct method. Part marks are shown in brackets for Sections B and C.

1. Large Numbers Around Us

1. b) 100
2. b) 10
3. b) 10,00,001
4. three crore five lakh forty thousand; 30,540,000 (1 mark for each)
5. 48,000 + 31,000 = 79,000 (1 mark for rounding, 1 mark for the estimate)
6. 1. More digits means bigger. 2. Same number of digits: compare from the left until two digits differ. Both have 8 digits (70456312 and 70456231); they first differ in the hundreds place, 3 against 2, so 7,04,56,312 is bigger. (1 mark for the digit-count step, 1 mark for comparing from the left, 1 mark for the correct answer)

2. Arithmetic Expressions

1. b) 4
2. a) 24
3. b) last
4. 6 - 2 = 4; 20 ÷ 4 = 5; 3 x 4 = 12; 5 + 12 = 17 (2 marks; 1 mark for brackets first)
5. (2 + 3) x 4 (2 marks)
6. Top: x. Its two branches: + joining 7 and 5, and - joining 9 and 4. 12 x 5 = 60. (1 mark for x at the top, 1 mark for the two branches, 1 mark for 60)

3. A Peek Beyond the Point

1. b) hundredths
2. a) 5
3. c) 32
4. 10011 (16 + 2 + 1) (2 marks; 1 mark for 16 + 2 + 1)
5. 26 (16 + 8 + 2) (2 marks; 1 mark for the place values)
6. 6 bits: 5 bits give only 32 codes, 6 bits give 64. 37 = 32 + 4 + 1 = 100101. (1 mark for 6, 1 mark for the reason, 1 mark for 100101)

4. Expressions using Letter-Numbers

1. b) 4s
2. a) 16
3. a) 7x
4. 5n + 1; step 20 = 101 (1 mark for each)
5. 8a + b (1 mark for 8a, 1 mark for b)
6. 6, 8, 10; n = 23 (1 mark for the table, 1 mark for working backwards (50 - 4 = 46, 46 ÷ 2), 1 mark for 23)

5. Parallel and Intersecting Lines

1. a) 40°
2. b) 180°
3. b) two
4. 108°, 72°, 108° (2 marks; 1 mark for 108°)
5. Yes. The heading is 0° for both (0°, then 60°, then back to 0°). (1 mark for yes, 1 mark for tracing the headings)
6. 4 are 115° and 4 are 65°. The angles come in two sizes, and each 115° angle sits on a straight line with a 65° angle (180° - 115°). (1 mark for each count, 1 mark for the explanation)

6. Number Play

1. b) 8
2. c) 16
3. a) removing the cases that break a clue
4. R and G; R and B; G and B (3 cases) (2 marks; 1 mark for two cases)
5. 45 and 45; both are multiples of 9 (1 mark for the answers, 1 mark for the pattern)
6. (10a + b) - (10b + a) = 9a - 9b = 9(a - b), which is a multiple of 9. (1 mark for 10a + b, 1 mark for the reverse 10b + a, 1 mark for 9(a - b))

7. A Tale of Three Intersecting Lines

1. b) 180°
2. b) 4, 5, 8
3. b) isosceles
4. 65°; acute (1 mark for each)
5. No: 5 + 5 = 10, which is not more than 10 (1 mark for no, 1 mark for the reason)
6. For example 10, 2, 3: 10 + 2 > 3 gives 'yes', but 2 + 3 < 10. Fix: if a + b > c and a + c > b and b + c > a then 'yes'. (1 mark for a failing test, 1 mark for explaining it, 1 mark for the fix)

8. Working with Fractions

1. b) 36
2. a) 2/5
3. a) 4/3
4. 6 ÷ 2/3 = 6 x 3/2 = 9 bottles (1 mark for the method, 1 mark for 9)
5. 23 + 5 = 28; 28 ÷ 4 = 7 (1 mark for each step)
6. ₹40 is 2/3 of the rest, so the rest was ₹60. ₹60 is half, so he started with ₹120. (1 mark for each step, 1 mark for ₹120)

AI 1. AI Domains and Applications

1. b) Computer Vision
2. b) Natural Language Processing
3. b) regression
4. (a) classification (b) clustering (1 mark for each)
5. 60; regression (1 mark for each)
6. For example 700 training, 150 validation, 150 test. Train on the first part, tune with the second, check on the third. Unseen test data shows how the model does on new photos. (1 mark for a sensible split, 1 mark for the job of each part, 1 mark for why)

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AI 2. AI in Industries

1. a) healthcare
2. b) travel time
3. b) the doctor
4. For example: personalised practice with instant feedback; translating or blocking spam (1 mark for each)
5. Route B, because it takes less time (1 mark for B, 1 mark for the reason)
6. Benefit: it checks many patients quickly. Risk: it can miss someone or be biased. Humans: a doctor reviews every suggestion before acting. (1 mark for each)

AI 3. Data Visualisation and Analysis

1. b) line graph
2. c) 360°
3. b) unstructured
4. 180°, 90°, 90° (2 marks; 1 mark for the method (count ÷ 60 x 360°))
5. It exaggerates a small difference. Start the axis at 0. (1 mark for each)
6. A line graph. It rises over the week, with a dip on Thursday. For example: label both axes and use equal steps. (1 mark for each)

AI 4. Ethics and AI Bias Awareness

1. a) the data
2. a) data bias
3. a) human in the loop
4. fruits 90%; vegetables 30% (1 mark for each)
5. Historical bias: the data repeats past unfair decisions (1 mark for the type, 1 mark for the reason)
6. Cause: few photos of children in the training data (data bias). Check: test accuracy for children and adults separately. Fix: add more children's photos and test again. (1 mark for each)

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Class record sheet

Class 7 ____ Section ____ Term ____. Write each score out of 10. You may photocopy this page.

Student	1	2	3	4	5	6	7	8	AI1	AI2	AI3	AI4	Avg

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